

Electronic Components Sector

Component Data Deposition: Trends in HDD shipments and major HDD components in Apr-Jun 2026

Techno Systems Research (TSR) released HDD shipment data for Apr-Jun (2Q 2026) on August 12. In 2Q 2026, HDD shipment volume was flat QoQ but grew 9% YoY, with nearline (NL) HDD shipments maintaining growth at 16% YoY. In addition, the capacity of HDD shipments increased significantly in 2Q 2026, rising 29% YoY, owing to higher NL HDD shipments and larger average capacity. In HDD components, HDD head shipment volume rose 19% YoY in 2Q 2026. HDD head shipments are expanding steadily, supported by growth in NL HDD shipments and an increase in the number of heads per unit. TDK is our top pick among HDD-related stocks in the electronic components sector. According to TSR data, TDK began supplying two captive manufacturers from 1Q 2026, and in 2Q, TDK expanded HDD head supply for NL applications. Regarding TDK, we believe that since captive manufacturers are restraining capacity expansion for HDD heads and shifting existing production capacity to HAMR, TDK should be able to gain share among captive manufacturers.

- 2Q 2026 HDD shipment data:** HDD shipment volume in 2Q 2026 was 33.0 million units (+9% YoY, flat QoQ), of which NL HDDs accounted for 19.1 million units (+16%, +2%). HDD shipment value was \$7,939 million (+47% YoY, +14% QoQ), of which NL HDDs accounted for \$6,912 million (+49%, +14%). The average HDD unit price was \$240.40 (+35% YoY, +14% QoQ), with the NL HDD average at \$362.80 (+29%, +12%). HDD shipment capacity was 504EB (+29% YoY, +6% QoQ), average capacity per drive was 15.3TB (+18%, +6%), and unit price per GB was \$0.0157 (+14%, +8%). In addition, the cost per GB for NL HDDs was \$0.014, and the GB cost gap between NAND flash and NL HDDs widened to 19.2x (4.8x in 2Q 2025, 12.0x in 1Q 2026), reaching a record high.
- 2Q 2026 HDD component trends:** HDD spindle motor shipment volume was 34.1 million units in 2Q 2026 (+12% YoY, flat QoQ). Nidec held a 60.1% share and MinebeaMitsumi had a 39.9% share. The HDD-SPM spindle ratio was 1.03 (1.00 in 2Q 2025, 1.03 in 1Q 2026). HGA (HDD head) shipment volume was 415 million units (+19% YoY, +3% QoQ), with 12.6 heads/HDD (11.5 heads/HDD in 2Q 2025, 12.2 heads/HDD in 1Q 2026), and 19.8 heads/NL HDD (18.6, 19.4). TDK's HGA shipment volume was 60.6 million units (+34% YoY, +15% QoQ), and its market share was 14.6% (13.0% in 2Q 2025, 13.1% in 1Q 2026). According to TSR data, supply to the two captive manufacturers that began in 1Q 2026 expanded further in 2Q.
- Also according to TSR data, Toshiba's HDD shipment value was \$873 million in 2Q 2026 (+55% YoY, +5% QoQ).

Japan Equity Research

Technology - Electronic Components

Akinori Kanemoto ^{AC}

(81-3) 6736 8628

akinori.kanemoto@jpmorgan.com

Ikki Shibata

(81-3) 6736 8641

ikki.shibata@jpmorgan.com

JPMorgan Securities Japan Co., Ltd.

Figure 1: Share price performance

Ticker	Company	Price (JPY)	Performance						Performance Ranking					
			-1W	-1M	-3M	-6M	-9M	-12M	-1W	-1M	-3M	-6M	-9M	-12M
6981	Murata Mfg.	7,538	1.6%	-16.9%	23.8%	131.3%	137.9%	210.7%	7	15	3	3	3	3
6976	Taiyo Yuden	10,265	3.3%	-13.7%	56.9%	152.6%	210.3%	260.0%	5	14	1	1	1	2
6762	TDK	3,114	3.4%	-2.8%	6.8%	40.8%	22.0%	56.2%	4	8	8	8	11	9
6971	Kyocera	3,790	-1.0%	4.2%	31.5%	43.9%	82.7%	96.7%	11	4	2	7	7	7
6963	Rohm	4,793	4.2%	-8.6%	19.8%	91.7%	122.6%	131.5%	2	11	5	5	6	5
6996	Nichicon	2,913	-2.4%	-9.0%	3.7%	49.2%	125.1%	124.9%	14	12	10	6	5	6
6997	Nippon Chemi-con	2,956	-8.8%	-19.1%	-7.0%	96.0%	132.2%	134.6%	16	16	15	4	4	4
6594	Nidec	2,734	-0.7%	3.4%	12.3%	17.2%	21.6%	-5.6%	10	5	6	11	12	15
6479	MinebeaMitsumi	3,753	0.5%	-9.9%	-4.1%	8.1%	28.7%	45.6%	9	13	13	12	10	10
4062	Ibiden	20,480	7.7%	9.6%	20.0%	137.0%	200.7%	517.1%	1	1	4	2	2	1
6806	Hirose Elec	27,020	-2.6%	0.6%	10.3%	30.8%	48.7%	40.0%	15	6	7	10	8	11
6807	JAE	2,301	-1.3%	-0.7%	-6.8%	-12.7%	-3.4%	-7.4%	12	7	14	16	16	16
6770	AlpsAlpine	2,188	3.4%	5.5%	-1.6%	1.1%	10.3%	30.2%	3	3	12	13	13	12
5334	Niterra	9,633	-1.4%	-6.0%	-0.7%	35.7%	43.0%	81.4%	13	10	11	9	9	8
6727	Wacom	807	2.2%	6.3%	5.6%	-6.5%	0.7%	9.2%	6	2	9	15	15	13
7915	NISSHA	1,275	0.5%	-3.1%	-12.3%	-2.4%	6.1%	-2.3%	8	9	16	14	14	14
Simple Average			0.5%	-3.8%	9.9%	50.9%	74.3%	107.7%						
	SOX	12,399	2.9%	0.4%	3.2%	52.4%	81.8%	110.4%						
	NK225	67,524	2.8%	0.4%	6.7%	18.6%	31.7%	56.0%						
	TOPIX	4,139	2.1%	3.3%	5.6%	8.4%	22.4%	33.9%						

Source: Bloomberg Finance L.P., J.P. Morgan.

Note: Share prices as of Aug 12, 2026. The darker the red, the higher the stock price performance ranking (the darker the blue, the lower the ranking).

Figure 2: Valuations of electronic component and HDD-related stocks

Ticker	Company	Share Price	Price Target	Return	Rating	Market Capitalization	P/E			EPS			EV/EBITDA			P/B		
		(JPY)	(JPY)				FY26E	FY27E	FY28E	FY26E	FY27E	FY28E	FY26E	FY27E	FY28E	FY26E	FY27E	FY28E
						(JPY mn)	(x)	(x)	(x)	(x)	(x)	(x)	(x)	(x)	(x)	(x)	(x)	(x)
6981	Murata Mfg.	7,538	15,200	101.6%	OW	16,540,254	42.56	26.34	17.62	177	286	428	29.53	19.39	13.39	4.89	4.32	3.64
6976	Taiyo Yuden	10,265	22,500	119.2%	OW	1,570,902	58.39	25.82	15.93	176	398	644	28.21	18.06	12.68	3.60	3.23	2.73
6762	TDK	3,114	5,500	76.6%	OW	6,335,039	26.31	20.66	16.54	118	151	188	12.10	10.04	8.31	2.53	2.35	2.15
6971	Kyocera	3,790	3,900	2.9%	N	5,538,751	33.20	29.54	26.43	114	128	143	18.25	15.84	13.84	1.49	1.57	1.65
6963	Rohm	4,793	8,000	66.9%	OW	2,008,706	38.22	28.05	22.45	125	171	213	21.84	17.81	12.90	2.52	2.40	2.14
6996	Nichicon	2,913	4,500	54.5%	OW	212,100	24.45	18.86	15.40	119	154	189	10.14	8.50	7.24	1.57	1.48	1.37
6997	Nippon Chemi-con	2,956	4,300	45.5%	N	87,897	18.78	13.16	9.33	157	225	317	10.96	8.95	7.15	1.46	1.30	1.13
6594	Nidec	2,734	1,800	-34.2%	UW	3,321,304	15.51	13.54	-	176	202	-	6.45	5.45	-	1.53	1.38	-
6479	MinebeaMitsumi	3,753	5,500	46.5%	OW	1,629,740	16.47	14.03	12.36	228	268	304	11.42	9.87	8.57	1.56	1.44	1.32
4062	Ibiden	20,480	28,500	39.2%	OW	6,329,643	75.37	56.26	41.85	272	364	489	30.72	24.50	19.82	9.21	8.02	6.81
6806	Hirose Elec	27,020	33,300	23.2%	OW	950,796	22.48	18.80	16.41	1,202	1,438	1,646	12.00	10.31	9.23	2.26	2.15	2.04
6807	JAЕ	2,301	2,280	-0.9%	N	164,227	23.90	15.00	12.01	96	153	192	4.90	4.13	3.61	1.05	1.01	0.96
6770	AlpsAlpine	2,188	2,480	13.3%	N	451,169	15.33	13.64	12.21	143	160	179	4.62	4.19	3.91	0.92	0.88	0.84
5334	Niterra	9,633	13,300	38.1%	OW	1,941,669	16.80	14.47	13.16	574	666	732	10.28	8.76	7.76	2.26	2.06	1.88
6727	Wacom	807	1,300	61.1%	OW	108,540	11.08	10.60	10.15	73	76	79	5.09	4.11	3.16	2.15	1.70	1.39
7915	NISSHA	1,275	1,460	14.5%	N	62,421	18.13	12.96	9.11	70	98	140	5.62	4.97	4.11	0.52	0.51	0.49
Simple Average		-	-	41.8%	-	2,953,322	28.56	20.73	16.73	239	309	392	13.88	10.93	9.05	2.47	2.24	2.04

Tiker	Company	Share Price	Price Target	Return	Rating	Market Capitalization	ROE			DPS			Dividend Yield			Dividend Payout Ratio		
		(JPY)	(JPY)				FY26E	FY27E	FY28E	FY26E	FY27E	FY28E	FY26E	FY27E	FY28E	FY26E	FY27E	FY28E
						(JPY mn)	(%)	(%)	(%)	(JPY)	(JPY)	(JPY)	(%)	(%)	(%)	(%)	(%)	(%)
6981	Murata Mfg.	7,538	15,200	101.6%	OW	16,540,254	11.6%	17.4%	22.4%	70.0	85.0	100.0	0.9%	1.1%	1.3%	39.5%	29.7%	23.4%
6976	Taiyo Yuden	10,265	22,500	119.2%	OW	1,570,902	6.9%	14.4%	20.3%	95.0	105.0	125.0	0.9%	1.0%	1.2%	54.0%	26.4%	19.4%
6762	TDK	3,114	5,500	76.6%	OW	6,335,039	9.9%	11.8%	13.6%	40.0	54.0	66.0	1.3%	1.7%	2.1%	33.8%	35.8%	35.0%
6971	Kyocera	3,790	3,900	2.9%	N	5,538,751	4.4%	5.2%	6.1%	56.0	56.0	56.0	1.5%	1.5%	1.5%	49.1%	43.6%	39.0%
6963	Rohm	4,793	8,000	66.9%	OW	2,008,706	7.8%	10.2%	11.5%	50.0	50.0	65.0	1.0%	1.0%	1.4%	39.9%	29.3%	30.5%
6996	Nichicon	2,913	4,500	54.5%	OW	212,100	6.5%	8.1%	9.2%	39.0	41.0	43.0	1.3%	1.4%	1.5%	32.7%	26.5%	22.7%
6997	Nippon Chemi-con	2,956	4,300	45.5%	N	87,897	7.2%	9.5%	12.0%	25.0	25.0	25.0	0.8%	0.8%	0.8%	15.9%	11.1%	7.9%
6594	Nidec	2,734	1,800	-34.2%	UW	3,321,304	10.3%	10.6%	-	0.0	0.0	-	0.0%	0.0%	-	0.0%	0.0%	-
6479	MinebeaMitsumi	3,753	5,500	46.5%	OW	1,629,740	9.8%	10.7%	11.1%	60.0	62.0	65.0	1.6%	1.7%	1.7%	26.3%	23.2%	21.4%
4062	Ibiden	20,480	28,500	39.2%	OW	6,329,643	13.7%	16.1%	18.6%	35.0	55.0	65.0	0.2%	0.3%	0.3%	12.9%	15.1%	13.3%
6806	Hirose Elec	27,020	33,300	23.2%	OW	950,796	10.1%	11.6%	12.6%	540.0	580.0	620.0	2.0%	2.1%	2.3%	44.9%	40.3%	37.7%
6807	JAЕ	2,301	2,280	-0.9%	N	164,227	4.4%	6.9%	8.2%	50.0	60.0	70.0	2.2%	2.6%	3.0%	51.9%	39.1%	36.6%
6770	AlpsAlpine	2,188	2,480	13.3%	N	451,169	6.4%	7.0%	7.5%	64.0	72.0	76.0	2.9%	3.3%	3.5%	44.8%	44.9%	42.4%
5334	Niterra	9,633	13,300	38.1%	OW	1,941,669	14.0%	14.9%	14.9%	220.0	250.0	280.0	2.3%	2.6%	2.9%	38.4%	37.6%	38.3%
6727	Wacom	807	1,300	61.1%	OW	108,540	22.3%	17.9%	15.1%	24.0	24.0	24.0	3.0%	3.0%	3.0%	32.9%	31.5%	30.2%
7915	NISSHA	1,275	1,460	14.5%	N	62,421	2.9%	4.0%	5.5%	50.0	50.0	50.0	3.9%	3.9%	3.9%	71.1%	50.8%	35.7%
Simple Average		-	-	41.8%	-	2,953,322	9.3%	11.0%	12.6%	-	-	-	1.6%	1.8%	2.0%	36.8%	30.3%	28.9%

Ticker	Company	Share Price	P/E Consensus(BBG)			P/B Consensus(BBG)		
		(LC)	FY26E	FY27E	FY28E	FY26E	FY27E	FY28E
			(x)	(x)	(x)	(x)	(x)	(x)
WDC U:WD		454	22.87	14.36	10.10	11.36	7.89	5.15
STX US Seagate		878	25.11	16.13	10.93	24.93	11.28	7.94
7741 JT HOYA		26,010	31.62	28.30	25.31	8.45	7.99	7.33
6504 JT Fuji electric		14,490	19.26	17.73	15.71	2.46	2.25	2.05
4004 JT Resonac HD		17,410	32.32	23.13	19.38	4.15	3.64	3.16
5991 JT NHK Spring		3,606	15.41	13.71	12.19	1.55	1.43	1.32

Source: Bloomberg Finance L.P., J.P. Morgan estimates.

Note: Share prices and Bloomberg (BBG) consensus are as of Aug 12, 2026.

Figure 3: HDD shipment volume, shipment value, average unit price, shipment capacity, average shipment capacity per unit, GB average unit price

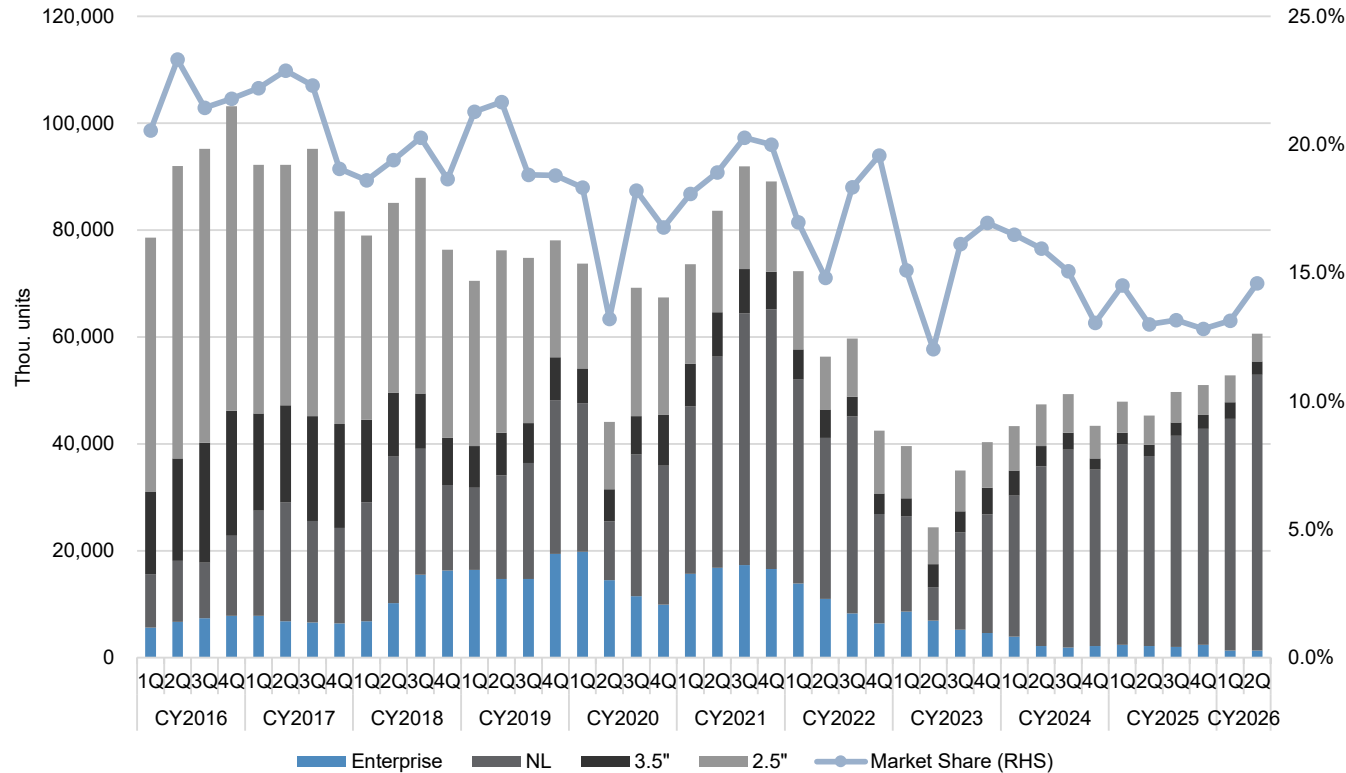
HDD shipment volume/revenue/EB shipment/Average capacity per drive (by formfactor)														
	CY2023				CY2024				CY2025				CY2026	
	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q
Shipment (mn units)														
Enterprise	1,410	1,600	1,350	1,400	1,500	1,020	0,950	0,940	0,820	0,810	0,730	0,790	0,760	0,600
yoy % chg.	-48.0%	-38.2%	-15.1%	-16.2%	6.4%	-36.3%	-29.6%	-32.9%	-45.3%	-20.6%	-23.2%	-16.0%	-7.3%	-25.9%
qoq % chg.	-15.6%	13.5%	-15.6%	3.7%	7.1%	-32.0%	-6.9%	-1.1%	-12.8%	-1.2%	-9.9%	8.2%	-3.8%	-21.1%
Near Line	11,400	9,150	8,950	9,900	12,300	13,810	15,820	16,530	14,610	16,470	18,140	18,400	18,660	19,050
yoy % chg.	-39.0%	-51.6%	-37.0%	-6.6%	7.9%	50.9%	76.8%	67.0%	18.8%	19.3%	14.7%	11.3%	27.7%	15.7%
qoq % chg.	7.5%	-19.7%	-2.2%	10.6%	24.2%	12.3%	14.6%	4.5%	-11.6%	12.7%	10.1%	1.4%	1.4%	2.1%
3.5" ATA	11,150	12,100	11,200	10,200	9,330	9,410	9,120	7,870	7,570	7,850	7,170	8,350	8,270	8,700
yoy % chg.	-32.6%	-4.0%	-14.1%	-20.3%	-16.3%	-22.2%	-18.6%	-22.8%	-18.9%	-16.6%	-21.4%	6.1%	9.2%	10.8%
qoq % chg.	-12.8%	8.5%	-7.4%	-8.9%	-8.5%	0.9%	-3.1%	-13.7%	-3.8%	3.7%	-8.7%	16.5%	-1.0%	5.2%
2.5"	9,500	8,430	7,120	7,400	6,610	6,090	6,210	6,400	5,700	5,270	5,570	5,500	5,320	4,680
yoy % chg.	-36.0%	-19.9%	-26.4%	-34.2%	-30.4%	-27.8%	-12.8%	-13.5%	-13.8%	-13.5%	-10.3%	-14.1%	-6.7%	-11.2%
qoq % chg.	-15.6%	-11.3%	-15.5%	3.9%	-10.7%	-7.9%	2.0%	3.1%	-10.9%	-7.5%	5.7%	-1.3%	-3.3%	-12.0%
Total	33,460	31,280	28,620	28,900	29,740	30,330	32,100	31,740	28,700	30,400	31,610	33,040	33,010	33,030
yoy % chg.	-37%	-30%	-26%	-20%	-11%	-3%	12%	10%	-3%	0%	-2%	4%	15%	9%
qoq % chg.	-8%	-7%	-9%	1%	3%	2%	6%	-1%	-10%	6%	4%	5%	0%	0%
Revenue (\$ mn)														
Enterprise	156.0	176.0	151.0	158.0	167.0	114.0	106.0	104.0	91.0	90.0	80.0	85.0	84.0	69.0
yoy % chg.	-48.2%	-38.7%	-14.2%	-14.6%	7.1%	-35.2%	-29.8%	-34.2%	-45.5%	-21.1%	-24.5%	-18.3%	-7.7%	-23.9%
qoq % chg.	-15.7%	12.8%	-14.2%	4.6%	5.7%	-31.7%	-7.0%	-1.9%	-12.5%	-1.1%	-11.1%	6.3%	-1.2%	-17.9%
Near Line	2,214.0	1,691.0	1,832.0	2,142.0	2,764.0	3,297.0	3,834.0	4,177.0	3,938.0	4,642.0	5,154.0	5,353.0	6,069.0	6,912.0
yoy % chg.	-35.5%	-52.8%	-36.9%	1.1%	24.8%	95.0%	109.3%	95.0%	42.5%	40.8%	34.4%	28.2%	54.1%	48.9%
qoq % chg.	4.5%	-23.6%	8.3%	16.9%	29.0%	19.3%	16.3%	8.9%	-5.7%	17.9%	11.0%	3.9%	13.4%	13.9%
3.5" ATA	626.0	686.0	639.0	590.0	533.0	532.0	520.0	453.0	432.0	449.0	414.0	545.0	547.0	669.0
yoy % chg.	-32.8%	-3.2%	-13.2%	-18.3%	-14.9%	-22.4%	-18.5%	-23.2%	-18.9%	-15.6%	-20.4%	20.3%	26.6%	49.0%
qoq % chg.	-13.3%	9.6%	-6.9%	-7.7%	-9.7%	-0.2%	-2.3%	-12.9%	-4.6%	3.9%	-7.6%	31.6%	0.4%	22.3%
2.5"	424.0	381.0	322.0	330.0	299.0	275.0	279.0	289.0	256.0	235.0	250.0	277.0	282.0	289.0
yoy % chg.	-38.1%	-20.6%	-26.0%	-32.7%	-29.5%	-27.8%	-13.4%	-14.7%	-14.4%	-14.5%	-10.4%	-4.2%	10.2%	23.0%
qoq % chg.	-15.9%	-10.1%	-15.5%	5.3%	-11.8%	-8.0%	1.5%	3.6%	-11.4%	-8.2%	6.4%	10.8%	1.8%	2.5%
Total	3,420.0	2,934.0	2,944.0	3,229.0	3,763.0	4,218.0	4,739.0	5,023.0	4,717.0	5,416.0	5,898.0	6,280.0	6,982.0	7,939.0
yoy % chg.	-36.1%	-42.0%	-30.7%	-8.5%	10.0%	43.8%	61.0%	55.6%	25.4%	28.4%	24.5%	24.6%	48.0%	46.6%
qoq % chg.	-3.1%	-14.2%	0.3%	9.7%	16.5%	12.1%	12.4%	6.0%	-6.1%	14.8%	8.9%	6.1%	11.5%	13.7%
ASP (\$)														
Enterprise	110.64	110.00	111.85	112.86	111.33	111.76	111.58	110.64	110.98	111.11	109.59	107.59	110.53	115.00
yoy % chg.	-0.4%	-0.7%	1.0%	1.9%	0.6%	1.6%	-0.2%	-2.0%	-0.3%	-0.6%	-1.8%	-2.8%	-0.4%	3.5%
qoq % chg.	-0.1%	-0.6%	1.7%	0.9%	-1.4%	0.4%	-0.2%	-0.8%	0.3%	0.1%	-1.4%	-1.8%	2.7%	4.0%
Near Line	194.21	184.81	204.69	216.36	224.72	238.74	242.35	252.69	269.54	281.85	284.12	290.92	325.24	362.83
yoy % chg.	5.7%	-2.5%	0.2%	8.3%	15.7%	29.2%	18.4%	16.8%	19.9%	18.1%	17.2%	15.1%	20.7%	28.7%
qoq % chg.	-2.8%	-4.8%	10.8%	5.7%	3.9%	6.2%	1.5%	4.3%	6.7%	4.6%	0.8%	2.4%	11.8%	11.6%
3.5" ATA	56.14	56.69	57.05	57.84	57.13	56.54	57.02	57.56	57.07	57.20	57.74	65.27	66.14	76.90
yoy % chg.	-0.4%	0.8%	1.1%	2.5%	1.8%	-0.3%	-0.1%	-0.5%	-0.1%	1.2%	1.3%	13.4%	15.9%	34.4%
qoq % chg.	-0.5%	1.0%	0.5%	1.4%	-1.2%	-1.0%	0.9%	1.0%	-0.9%	0.2%	0.9%	13.0%	1.3%	16.3%
2.5"	44.63	45.20	45.22	45.91	45.23	45.16	44.93	45.16	44.91	44.59	44.88	50.36	53.01	61.75
yoy % chg.	-3.2%	-0.9%	0.6%	2.3%	1.4%	-0.1%	-0.7%	-1.4%	-0.7%	-1.2%	-0.1%	11.5%	18.0%	38.5%
qoq % chg.	-0.4%	1.3%	0.1%	1.3%	-1.3%	-0.2%	-0.5%	0.5%	-0.5%	-0.7%	0.7%	12.2%	5.2%	16.5%
Total	102.21	93.80	102.87	111.73	126.53	139.07	147.63	158.25	164.36	178.16	186.59	189.47	211.51	240.38
yoy % chg.	0.8%	-17.2%	-6.8%	15.0%	23.8%	48.3%	43.5%	41.6%	29.9%	28.1%	26.4%	19.7%	28.7%	34.9%
qoq % chg.	5.2%	-8.2%	9.7%	8.6%	13.2%	9.9%	6.2%	7.2%	3.9%	8.4%	4.7%	1.5%	11.6%	13.6%
HDD Capacity shipment and average capacity per drive														
	CY2023				CY2024				CY2025				CY2026	
	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q
Capacity Shipment (EB)														
Enterprise	236.0	189.8	196.1	214.3	264.6	289.4	340.3	362.3	346.6	391.8	433.4	457.0	477.1	504.3
yoy % chg.	-28%	-42%	-31%	-25%	12%	52%	74%	69%	31%	35%	27%	26%	38%	29%
qoq % chg.	-17%	-20%	3%	9%	23%	9%	18%	6%	-4%	13%	11%	5%	4%	6%
Seagate	118.7	91.2	89.6	95.1	99.1	114.2	137.5	150.8	143.6	162.5	181.7	190.1	199.0	218.0
yoy % chg.	-23%	-41%	-24%	-20%	-17%	25%	53%	59%	45%	42%	32%	26%	39%	34%
qoq % chg.	0%	-23%	-2%	6%	4%	15%	20%	10%	-5%	13%	12%	5%	5%	10%
WDC	96.5	78.8	79.1	90.3	127.3	142.6	163.8	172.5	162.0	187.0	205.0	215.0	222.0	231.0
yoy % chg.	-27%	-41%	-39%	-31%	32%	81%	107%	91%	27%	31%	25%	25%	37%	24%
qoq % chg.	-26%	-18%	0%	14%	41%	12%	15%	5%	-6%	15%	10%	5%	3%	4%
Toshiba	20.8	19.8	27.4	28.9	38.2	32.6	39.0	39.0	41.0	42.3	46.7	51.9	56.1	55.3
yoy % chg.	-51%	-45%	-26%	-22%	84%	65%	43%	35%	7%	30%	20%	33%	37%	31%
qoq % chg.	-44%	-5%	38%	5%	32%	-15%	20%	0%	5%	3%	10%	11%	8%	-1%
Average Capacity per Drive (GB)	7,053.2	6,067.8	6,851.9	7,415.2	8,879.8	9,541.1	10,606.0	11,413.2	12,075.9	12,887.7	13,710.9	13,831.7	14,453.2	15,267.9
yoy % chg.	13%	-17%	-8%	-6%	26%	57%	55%	54%	36%	35%	29%	21%	20%	18%
qoq % chg.	-10%	-14%	13%	8%	20%	7%	11%	8%	6%	7%	6%	1%	4%	6%
Seagate	8,186.2	6,377.6	7,466.7	8,198.3	8,693.0	9,284.6	10,617.8	11,600.0	12,596.5	13,000.0	14,571.0	14,567.0	15,237.4	16,452.8
yoy % chg.	22%	-18%	0%	7%	6%	46%	42%	41%	45%	40%	37%	26%	21%	27%
qoq % chg.	7%	-22%	17%	10%	6%	7%	14%	9%	9%	3%	12%	0%	5%	8%
WDC	7,658.7	6,678.0	7,605.8	8,361.1	10,880.3	11,785.1	12,223.9	12,777.8	13,388.4	14,496.1	14,748.2	15,003.5	15,416.7	15,876.3
yoy % chg.	14%	-18%	-14%	-17%	42%	76%	61%	53%	23%	23%	21%	17%	15%	10%
qoq % chg.	-24%	-13%	14%	10%	30%	8%	4%	5%	5%	8%	2%	2%	3%	3%
Toshiba	3,270.4	3,822.4	4,405.1	4,446.2	5,704.0	5,494.4	6,802.8	7,434.4	7,880.6	8,457.2	8,912.2	9,169.6	10,108.1	10,573.6
yoy % chg.	-23%	-12%	-5%	-4%	74%	44%	54%	67%	38%	54%	31%	23%	28%	25%
qoq % chg.	-30%	17%	15%	1%	28%	4%	24%	9%	6%	7%	5%	3%	10%	5%
Per GB price (\$)	0.0145	0.0155	0.0150	0.0151	0.0142	0.0146	0.0139	0.0138	0.0136	0.0138	0.0136	0.0137	0.0146	0.0151
yoy % chg.	-11%	-1%	1%	22%	-2%	-7%	-8%	-4%	-5%	-2%	-1%	8%	14%	14%
qoq % chg.	-17%	7%	-3%	0%	-6%	2%	-4%	0%	-2%	-2%	-2%	1%	7%	8%
Seagate	0.0135	0.0152	0.0145	0.0146	0.0149	0.0151	0.0146	0.0144	0.0139	0.0140	0.0132	0.0136	0.0144	0.0155
yoy % chg.	-19%	3%	-4%	3%	10%	0%	1%	-1%	-6%	-7%	-9%	-5%	3%	10%
qoq % chg.	-4%	-12%	-5%	1%	2%	1%	-4%	-1%	-3%	1%	-6%	3%	6%	7%
WDC	0.0142	0.0148	0.0147	0.0146	0.0133	0.0138	0.0132	0.0135	0.0134	0.0137	0.0137	0.0137	0.0148	0.0160
yoy % chg.	-5%	1%	4%	49%	-6%	-7%	-10%	-8%	0%	-1%	4%	2%	11%	16%
qoq % chg.	45%	4%	-1%	-1%	-9%	4%	-4%	2%	-1%	3%	0%	0%	8%	8%
Toshiba	0.0213													

Figure 4: HDD shipment volume, market share, and shipment value by manufacturer

HDD shipment by formfactor and manufacture																
	CY2023				CY2024				CY2025				CY2026			
	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q		
Thou. Units																
Enterprise	1,410	1,600	1,350	1,400	1,500	1,020	950	940	820	810	730	790	760	600	8,560	5,760
Seagate	750	1,000	750	700	700	600	600	550	450	450	400	400	390	400	5,700	3,200
WD	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Toshiba/Fujitsu	660	600	600	700	800	420	350	390	370	360	330	390	370	200	2,860	2,560
NL	11,400	9,150	8,950	9,900	12,300	13,810	15,820	16,530	14,610	16,470	18,140	18,400	18,660	19,050	62,370	39,400
Seagate	5,750	4,400	4,250	4,500	4,800	5,600	6,500	6,850	5,850	6,600	7,500	7,300	7,400	7,650	29,100	18,900
WD	4,550	3,700	3,300	3,900	5,600	6,400	7,200	7,650	6,700	7,800	8,400	8,600	8,750	8,950	24,850	15,450
Toshiba/Fujitsu	1,100	1,050	1,400	1,500	1,900	1,810	2,120	2,030	2,060	2,070	2,240	2,500	2,510	2,450	8,420	5,050
3.5"ATA	11,150	12,100	11,200	10,200	9,400	9,410	9,120	7,870	7,570	7,850	7,170	8,350	8,270	8,700	54,980	44,650
Seagate	4,500	5,500	4,900	3,900	3,800	4,000	3,850	3,400	3,200	3,500	2,600	3,400	3,400	3,650	22,500	18,800
WD	5,150	5,300	4,500	4,500	4,000	3,900	4,100	3,750	3,500	3,600	3,800	4,000	3,800	4,250	24,350	19,450
Toshiba/Fujitsu	1,500	1,300	1,800	1,800	1,600	1,510	1,170	720	870	750	770	950	1,070	800	8,130	6,400
2.5"	9,500	8,430	7,120	7,400	6,600	6,090	6,200	6,400	5,700	5,270	5,570	5,500	5,320	4,680	46,310	32,450
Seagate	3,500	3,400	2,100	2,500	2,100	2,100	2,000	2,200	1,900	1,950	1,970	1,950	1,870	1,550	16,800	11,500
WD	2,900	2,800	2,600	2,400	2,100	1,800	2,100	2,100	1,900	1,500	1,700	1,730	1,850	1,350	14,650	10,700
Toshiba/Fujitsu	3,100	2,230	2,420	2,500	2,400	2,190	2,100	2,100	1,900	1,820	1,900	1,820	1,600	1,780	14,860	10,250
Total	33,460	31,280	28,620	28,900	29,800	30,330	32,090	31,740	28,700	30,400	31,610	33,040	33,010	33,030	172,220	122,260
Seagate	14,500	14,300	12,000	11,600	11,400	12,300	12,950	13,000	11,400	12,500	12,470	13,050	13,060	13,250	74,100	52,400
WD	12,600	11,800	10,400	10,800	11,700	12,100	13,400	13,500	12,100	12,900	13,900	14,330	14,400	14,550	63,850	45,600
HGST	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	0
Samsung	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0	0
Toshiba/Fujitsu	6,360	5,180	6,220	6,500	6,700	5,930	5,740	5,240	5,200	5,000	5,240	5,660	5,550	5,230	34,270	24,260
Market share in shipment basis																
	CY2023				CY2024				CY2025				CY2026			
Market share	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q		
Enterprise	53.2%	62.5%	55.6%	50.0%	46.7%	58.8%	63.2%	58.5%	54.9%	55.6%	54.8%	50.6%	51.3%	66.7%	66.6%	55.6%
Seagate	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%
WD	46.8%	37.5%	44.4%	50.0%	53.3%	41.2%	36.8%	41.5%	45.1%	44.4%	45.2%	49.4%	48.7%	33.3%	33.4%	44.4%
Toshiba/Fujitsu	50.4%	48.1%	47.5%	45.5%	39.0%	40.6%	41.1%	41.4%	40.0%	40.1%	41.3%	39.7%	39.7%	40.2%	46.7%	48.0%
NL	39.9%	40.4%	36.9%	39.4%	45.5%	46.3%	45.5%	46.3%	45.9%	47.4%	46.3%	46.7%	46.9%	47.0%	39.8%	39.2%
Seagate	9.6%	11.5%	15.6%	15.2%	15.4%	13.1%	13.4%	12.3%	14.1%	12.6%	12.3%	13.6%	13.5%	12.9%	13.5%	12.8%
WD	40.4%	45.5%	43.8%	38.2%	40.4%	42.5%	42.2%	43.2%	42.3%	44.6%	36.3%	40.7%	41.1%	42.0%	40.9%	42.1%
Toshiba/Fujitsu	46.2%	43.8%	40.2%	44.1%	42.6%	41.4%	45.0%	47.6%	46.2%	45.9%	53.0%	47.9%	45.9%	48.9%	44.3%	43.6%
3.5"ATA	13.5%	10.7%	16.1%	17.6%	17.0%	16.0%	12.8%	9.1%	11.5%	9.6%	10.7%	11.4%	12.9%	9.2%	14.8%	14.3%
Seagate	36.8%	40.3%	29.5%	33.8%	31.8%	34.5%	32.3%	34.4%	33.3%	37.0%	35.4%	35.5%	35.2%	33.1%	36.3%	35.4%
WD	30.5%	33.2%	36.5%	32.4%	31.8%	29.6%	33.9%	32.8%	33.3%	28.5%	30.5%	31.5%	34.8%	28.8%	31.6%	33.0%
Toshiba/Fujitsu	32.6%	26.5%	34.0%	33.8%	36.4%	36.0%	33.9%	32.8%	33.3%	34.4%	34.1%	33.1%	30.1%	38.0%	32.1%	31.6%
Total	43.3%	45.7%	41.9%	40.1%	38.3%	40.6%	40.4%	41.0%	39.7%	41.1%	39.4%	39.5%	39.6%	40.1%	43.0%	42.9%
Seagate	37.7%	37.7%	36.3%	37.4%	39.3%	39.9%	41.8%	42.5%	42.2%	42.4%	44.0%	43.4%	43.6%	44.1%	37.1%	37.3%
WD	19.0%	16.6%	21.7%	22.5%	22.5%	19.6%	17.9%	16.5%	18.1%	16.4%	16.6%	17.1%	16.8%	15.8%	19.9%	19.8%
Toshiba/Fujitsu																
HDD revenue (by formfactor and manufacture)																
	CY2023				CY2024				CY2025				CY2026			
	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q	3Q	4Q	1Q	2Q		
\$ mn																
Enterprise	156	176	151	158	167	114	106	104	91	90	80	85	84	69	949	641
Seagate	83	110	84	79	78	67	67	61	50	50	44	44	43	49	634	356
WD	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Toshiba/Fujitsu	73	66	67	79	89	47	39	43	41	40	36	41	41	20	315	285
NL	2,214	1,691	1,832	2,142	2,764	3,297	3,834	4,177	3,938	4,642	5,154	5,353	6,069	6,912	12,034	7,879
Seagate	1,103	804	826	953	1,076	1,327	1,618	1,806	1,678	1,937	2,114	2,225	2,500	2,933	5,696	3,686
WD	920	710	757	914	1,344	1,644	1,821	1,991	1,860	2,279	2,510	2,578	2,912	3,273	4,867	3,301
Toshiba/Fujitsu	191	177	249	275	344	326	395	380	400	426	530	550	657	706	1,471	892
3.5"ATA	626	686	639	590	533	532	520	453	432	449	414	545	547	669	3,099	2,541
Seagate	254	312	285	231	222	232	223	197	185	202	151	219	220	283	1,280	1,082
WD	310	320	278	282	246	237	248	226	211	216	221	275	263	331	1,483	1,190
Toshiba/Fujitsu	62	54	76	77	65	63	49	30	36	31	42	51	64	55	336	269
2.5"	424	381	322	339	299	275	279	289	256	235	250	277	282	289	2,104	1,466
Seagate	164	159	100	121	101	101	96	105	90	92	91	104	101	105	800	544
WD	143	138	130	122	106	91	96	105	95	75	87	103	110	92	735	533
Toshiba/Fujitsu	117	84	92	96	92	83	87	79	71	68	72	70	71	92	569	389
Total	3,420	2,934	2,944	3,229	3,763	4,218	4,739	5,023	4,717	5,416	5,898	6,260	6,982	7,939	18,186	12,527
Seagate	1,604	1,385	1,295	1,384	1,477	1,727	2,004	2,169	2,003	2,281	2,400	2,592	2,864	3,370	8,410	5,668
WD	1,373	1,168	1,165	1,318	1,696	1,972	2,165	2,322	2,166	2,570	2,818	2,956	3,285	3,696	7,085	5,024
Toshiba/Fujitsu	443	381	484	527	590	519	570	532	548	565	680	712	833	873	2,691	1,835

Source: TSR, J.P. Morgan.

Figure 5: HDD head numbers, TDK's HDD head shipment volume, market share



Source: TSR, J.P. Morgan.

Companies Discussed in This Report (all prices in this report as of market close on 13 August 2026, unless otherwise indicated)
TDK (6762)(6762.T/¥3,235/OW)

Analyst Certification: The Research Analyst(s) denoted by an “AC” on the cover of this report certifies (or, where multiple Research Analysts are primarily responsible for this report, the Research Analyst denoted by an “AC” on the cover or within the document individually certifies, with respect to each security or issuer that the Research Analyst covers in this research) that: (1) all of the views expressed in this report accurately reflect the Research Analyst’s personal views about any and all of the subject securities or issuers; and (2) no part of any of the Research Analyst’s compensation was, is, or will be directly or indirectly related to the specific recommendations or views expressed by the Research Analyst(s) in this report. For all Korea-based Research Analysts listed on the front cover, if applicable, they also certify, as per KOFIA requirements, that the Research Analyst’s analysis was made in good faith and that the views reflect the Research Analyst’s own opinion, without undue influence or intervention.

All authors named within this report are Research Analysts who produce independent research unless otherwise specified. In Europe, Sector Specialists (Sales and Trading) may be shown on this report as contacts but are not authors of the report or part of the Research Department.

Important Disclosures

- **Market Maker/ Liquidity Provider:** J.P. Morgan is a market maker and/or liquidity provider in the financial instruments of/related to TDK (6762) or related entities.
- **Client:** J.P. Morgan currently has, or had within the past 12 months, the following entity(ies) as clients: TDK (6762) or related entities.
- **Client/Non-Investment Banking, Securities-Related:** J.P. Morgan currently has, or had within the past 12 months, the following entity(ies) as clients, and the services provided were non-investment-banking, securities-related: TDK (6762) or related entities.
- **Potential Investment Banking Compensation:** J.P. Morgan expects to receive, or intends to seek, compensation for investment banking services in the next three months from TDK (6762) or related entities.
- **Non-Investment Banking Compensation Received:** J.P. Morgan has received compensation in the past 12 months for products or services other than investment banking from TDK (6762) or related entities.
- **Debt Position:** J.P. Morgan may hold a position in the debt securities of TDK (6762) or related entities, if any.

Company-Specific Disclosures: J.P. Morgan does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision. Important disclosures, including price charts and credit opinion history tables (if applicable), are available for compendium reports and all J.P. Morgan-covered companies, and certain non-covered companies, by visiting <https://www.jpmm.com/research/disclosures>, calling 1-800-477-0406, or e-mailing research.disclosure.inquiries@jpmorgan.com with your request.

TDK (6762) (6762.T, 6762 JP) Price Chart



Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
Initiated coverage Jan 24, 2002. All share prices are as of market close on the previous business day.
Break in coverage Jul 04, 2024 - Jul 22, 2025.

The chart(s) show J.P. Morgan's continuing coverage of the stocks; the current analysts may or may not have covered it over the entire period. J.P. Morgan ratings or designations: OW = Overweight, N= Neutral, UW = Underweight, NR = Not Rated

Explanation of Equity Research Ratings, Designations and Analyst(s) Coverage Universe:

J.P. Morgan uses the following rating system: Overweight (over the duration of the price target indicated in this report, we expect this stock will outperform the average total return of the stocks in the Research Analyst's, or the Research Analyst's team's, coverage universe); Neutral (over the duration of the price target indicated in this report, we expect this stock will perform in line with the average total return of the stocks in the Research Analyst's, or the Research Analyst's team's, coverage universe); and Underweight (over the duration of the price target indicated in this report, we expect this stock will underperform the average total return of the stocks in the Research Analyst's, or the Research Analyst's team's, coverage universe. NR is Not Rated. In this case, J.P. Morgan has removed the rating and, if applicable, the price target, for this stock because of either a lack of a sufficient fundamental basis or for legal, regulatory or policy reasons. The previous rating and, if applicable, the price target, no longer should be relied upon. An NR designation is not a recommendation or a rating. Some stocks under coverage have a rating but no price target; in these cases, we expect the stock will outperform/perform in line/underperform the average total return of the stocks in the Research Analyst's, or the Research Analyst's team's, coverage universe of the relevant duration of the region. In our Asia (ex-Australia and ex-India) and U.K. small- and mid-cap Equity Research, each stock's expected total return is compared to the expected total return of a benchmark country market index, not to those Research Analysts' coverage universe. If it does not appear in the Important Disclosures section of this report, the certifying Research Analyst's coverage universe can be found on J.P. Morgan's Research website, <https://www.jpmorganmarkets.com>.

Coverage Universe: Kanemoto, Akinori : Alps Alpine (6770) (6770.T), Hirose Electric (6806) (6806.T), Ibiben (4062) (4062.T), Japan Aviation Electronics (6807) (6807.T), Kyocera (6971) (6971.T), MinebeaMitsumi (6479) (6479.T), Murata Manufacturing (6981) (6981.T), NISSHA (7915) (7915.T), Nichicon (6996) (6996.T), Nidec (6594) (6594.T), Nippon Chemi-Con (6997) (6997.T), Niterra (5334) (5334.T), Rohm (6963) (6963.T), TDK (6762) (6762.T), Taiyo Yuden (6976) (6976.T), Wacom (6727) (6727.T)

J.P. Morgan Equity Research Ratings Distribution, as of July 04, 2026

	Overweight (buy)	Neutral (hold)	Underweight (sell)
J.P. Morgan Global Equity Research Coverage*	53%	36%	12%
IB clients**	83%	80%	73%
JPMS Equity Research Coverage*	51%	37%	12%
IB clients**	95%	92%	87%

*Please note that the percentages may not add to 100% because of rounding.

**Percentage of subject companies within each of the "buy," "hold" and "sell" categories for which J.P. Morgan has provided investment banking services within the previous 12 months.

For purposes of FINRA ratings distribution rules only, our Overweight rating falls into a buy rating category; our Neutral rating falls into a hold rating category; and our Underweight rating falls into a sell rating category. Please note that stocks with an NR designation are not included in the table above. This information is current as of the end of the most recent calendar quarter.

Equity Valuation and Risks: For valuation methodology and risks associated with covered companies or price targets for covered companies, please see the most recent company-specific research report at <http://www.jpmorganmarkets.com>, contact the primary analyst or your J.P. Morgan representative, or email research.disclosure.inquiries@jpmorgan.com. For material information about the proprietary models used, please see the Summary of Financials in company-specific research reports and the Company Tearsheets, which are available to download on the company pages of our client website, <http://www.jpmorganmarkets.com>. This report also sets out within it the material underlying assumptions used.

History of Investment Recommendations:

A history of J.P. Morgan investment recommendations disseminated during the preceding 12 months can be accessed on the Research & Commentary page of <http://www.jpmorganmarkets.com> where you can also search by analyst name, sector or financial instrument.

Analysts' Compensation: The research analysts responsible for the preparation of this report receive compensation based upon various factors, including the quality and accuracy of research, client feedback, competitive factors, and overall firm revenues.

Registration of non-US Analysts: Unless otherwise noted, the non-US analysts listed on the front of this report are employees of non-US affiliates of J.P. Morgan Securities LLC, may not be registered as research analysts under FINRA rules, may not be associated persons of J.P. Morgan Securities LLC, and may not be subject to FINRA Rule 2241 or 2242 restrictions on communications with covered companies, public appearances, and trading securities held by a research analyst account.

Other Disclosures

J.P. Morgan is a marketing name for investment banking businesses of JPMorgan Chase & Co. and its subsidiaries and affiliates worldwide.

UK MIFID FICC research unbundling exemption: UK clients should refer to [UK MIFID Research Unbundling exemption](#) for details of J.P. Morgan's implementation of the FICC research exemption and guidance on relevant FICC research categorisation.

All research material made available to clients are simultaneously available on our client website, J.P. Morgan Markets, unless specifically permitted by relevant laws. Not all research content is redistributed, e-mailed or made available to third-party aggregators. For all research material available on a particular stock, please contact your sales representative.

Any long form nomenclature for references to China; Hong Kong; Taiwan; and Macau within this research material are Mainland China; Hong Kong SAR (China); Taiwan (China); and Macau SAR (China).

J.P. Morgan Research may, from time to time, write on issuers or securities targeted by economic or financial sanctions imposed or administered by the governmental authorities of the U.S., EU, UK or other relevant jurisdictions (Sanctioned Securities). Nothing in this report is intended to be read or construed as encouraging, facilitating, promoting or otherwise approving investment or dealing in such Sanctioned Securities. Clients should be aware of their own legal and compliance obligations when making investment decisions.

Any digital or crypto assets discussed in this research report are subject to a rapidly changing regulatory landscape. For relevant regulatory advisories on crypto assets, including bitcoin and ether, please see <https://www.jpmorgan.com/disclosures/cryptoasset-disclosure>.

The author(s) of this research report may not be licensed to carry on regulated activities in your jurisdiction and, if not licensed, do not hold themselves out as being able to do so.

Exchange-Traded Funds (ETFs): J.P. Morgan Securities LLC (“JPMS”) acts as authorized participant for substantially all U.S.-listed ETFs. To the extent that any ETFs are mentioned in this report, JPMS may earn commissions and transaction-based compensation in connection with the distribution of those ETF shares and may earn fees for performing other trade-related services, such as securities lending to short sellers of the ETF shares. JPMS may also perform services for the ETFs themselves, including acting as a broker or dealer to the ETFs. In addition, affiliates of JPMS may perform services for the ETFs, including trust, custodial, administration, lending, index calculation and/or maintenance and other services.

Options and Futures related research: If the information contained herein regards options- or futures-related research, such information is available only to persons who have received the proper options or futures risk disclosure documents. Please contact your J.P. Morgan Representative or visit <https://www.theocc.com/components/docs/riskstoc.pdf> for a copy of the Option Clearing Corporation's Characteristics and Risks of Standardized Options or https://www.finra.org/sites/default/files/2020-08/Security_Futures_Risk_Disclosure_Statement_2020.pdf for a copy of the Security Futures Risk Disclosure Statement.

Changes to Interbank Offered Rates (IBORs) and other benchmark rates: Certain interest rate benchmarks are, or may in the future become, subject to ongoing international, national and other regulatory guidance, reform and proposals for reform. For more information, please consult: https://www.jpmorgan.com/global/disclosures/interbank_offered_rates

Notification for Credit Ratings: If this material includes credit ratings, such credit ratings provided by Japan Credit Rating Agency, Ltd. (JCR) and Rating and Investment Information, Inc. (R&I), are credit ratings provided by Registered Credit Rating Agencies (credit rating agencies registered under the Financial Instruments and Exchange Law of Japan (FIEL)). With respect to credit ratings that are provided by credit rating agencies other than JCR and R&I and have no stipulation that such credit ratings are provided by Registered Credit Rating Agencies, this means that such credit ratings are Non Registered Ratings (credit ratings provided by credit rating agencies not registered under the FIEL). Among the Non Registered Ratings, with respect to those credit ratings provided by S&P Global Ratings (S&P), Moody's Investors Service (Moody's), or Fitch Ratings (Fitch), prior to making investment decision based on such Non Registered Ratings, please carefully read the “Explanation Letter regarding Non Registered Ratings” for the corresponding credit rating agency, which we separately have sent or will send.

Private Bank Clients: Where you are receiving research as a client of the private banking businesses offered by JPMorgan Chase & Co. and its subsidiaries (“J.P. Morgan Private Bank”), research is provided to you by J.P. Morgan Private Bank and not by any other division of J.P. Morgan, including, but not limited to, the J.P. Morgan Corporate and Investment Bank and its Global Research division.

Legal entity responsible for the production and distribution of research: The legal entity identified below the name of the Reg AC Research Analyst who authored this material is the legal entity responsible for the production of this research. Where multiple Reg AC Research Analysts authored this material with different legal entities identified below their names, these legal entities are jointly responsible for the production of this research. Where more than one legal entity is listed under an analyst's name, the first legal entity is responsible for the production unless stated otherwise. Research Analysts from various J.P. Morgan affiliates may have contributed to the production of this material but may not be licensed to carry out regulated activities in your jurisdiction (and do not hold themselves out as being able to do so). Unless otherwise stated below in the legal entity disclosures, this material has been distributed by the legal entity responsible for production, or where more than one legal entity is listed under the analyst's name, the first legal entity will be responsible for distribution. If you have any queries, please contact the relevant Research Analyst in your jurisdiction or the entity in your jurisdiction that has distributed this research material.

Legal Entities Disclosures and Country-/Region-Specific Disclosures:

Argentina: JPMorgan Chase Bank N.A Sucursal Buenos Aires is regulated by Banco Central de la República Argentina (“BCRA”- Central Bank of Argentina) and Comisión Nacional de Valores (“CNV”- Argentinian Securities Commission - ALYC y AN Integral N°51).

Australia: J.P. Morgan Securities Australia Limited (“JPMSAL”) (ABN 61 003 245 234/AFS Licence No: 238066) is regulated by the Australian Securities and Investments Commission and is a Market Participant of ASX Limited, a Clearing and Settlement Participant of ASX Clear Pty Limited and a Clearing Participant of ASX Clear (Futures) Pty Limited. This material is issued and distributed in Australia by or on behalf of JPMSAL only to “wholesale clients” (as defined in section 761G of the Corporations Act 2001). A list of all financial products covered can be found by visiting <https://www.jpmm.com/research/disclosures>. J.P. Morgan seeks to cover companies of relevance to the domestic and international investor base across all Global Industry Classification Standard (GICS) sectors, as well as across a range of market capitalisation sizes. If applicable, in the course of conducting public side due diligence on the subject company(ies), the Research Analyst team may at times

perform such diligence through corporate engagements such as site visits, discussions with company representatives, management presentations, etc. Research issued by JPMSAL has been prepared in accordance with J.P. Morgan Australia's Research Independence Policy which can be found at the following link: [J.P. Morgan Australia - Research Independence Policy](#).

Brazil: Banco J.P. Morgan S.A. is regulated by the Comissao de Valores Mobiliarios (CVM) and by the Central Bank of Brazil. Ombudsman J.P. Morgan: 0800-7700847 / 0800-7700810 (For Hearing Impaired) / ouvidoria.jp.morgan@jpmchase.com.

Canada: J.P. Morgan Securities Canada Inc. is a registered investment dealer, regulated by the Canadian Investment Regulatory Organization and the Ontario Securities Commission and is the participating member on Canadian exchanges. This material is distributed in Canada by or on behalf of J.P. Morgan Securities Canada Inc.

Chile: Inversiones J.P. Morgan Limitada is an unregulated entity incorporated in Chile.

China: J.P. Morgan Securities (China) Company Limited has been approved by CSRC to conduct the securities investment consultancy business.

Colombia: Banco J.P. Morgan Colombia S.A. is supervised by the Superintendencia Financiera de Colombia (SFC). Any reference in this material to products or services offered abroad by entities other than the Bank in Colombia is included exclusively for descriptive purposes. Such references do not constitute, and should not be construed as, promotional activity or the provision of financial products or services within Colombian territory, as defined under applicable Colombian regulation.

Dubai International Financial Centre (DIFC): JPMorgan Chase Bank, N.A., Dubai Branch is regulated by the Dubai Financial Services Authority (DFSA) and its registered address is Dubai International Financial Centre - The Gate, West Wing, Level 3 and 9 PO Box 506551, Dubai, UAE. This material has been distributed by JP Morgan Chase Bank, N.A., Dubai Branch to persons regarded as professional clients or market counterparties as defined under the DFSA rules.

European Economic Area (EEA): Unless specified to the contrary, research is distributed in the EEA by J.P. Morgan SE ("JPM SE"), which is authorised as a credit institution by the Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and jointly supervised by the BaFin, the German Central Bank (Deutsche Bundesbank) and the European Central Bank (ECB). JPM SE is a company headquartered in Frankfurt with registered address at TaunusTurm, Taunustor 1, Frankfurt am Main, 60310, Germany. The material has been distributed in the EEA to persons regarded as professional investors (or equivalent) pursuant to Art. 4 para. 1 no. 10 and Annex II of MiFID II and its respective implementation in their home jurisdictions ("EEA professional investors"). This material must not be acted on or relied on by persons who are not EEA professional investors. Any investment or investment activity to which this material relates is only available to EEA relevant persons and will be engaged in only with EEA relevant persons.

Hong Kong: J.P. Morgan Securities (Asia Pacific) Limited (CE number AAJ321) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission in Hong Kong, and J.P. Morgan Broking (Hong Kong) Limited (CE number AAB027) is regulated by the Securities and Futures Commission in Hong Kong. JP Morgan Chase Bank, N.A., Hong Kong Branch (CE Number AAL996) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission, is organized under the laws of the United States with limited liability. Where the distribution of this material is a regulated activity in Hong Kong, the material is distributed in Hong Kong by or through J.P. Morgan Securities (Asia Pacific) Limited and/or J.P. Morgan Broking (Hong Kong) Limited.

India: J.P. Morgan India Private Limited (Corporate Identity Number - U67120MH1992FTC068724), having its registered office at J.P. Morgan Tower, Off. C.S.T. Road, Kalina, Santacruz - East, Mumbai - 400098, is registered with the Securities and Exchange Board of India (SEBI) as a 'Research Analyst' having registration number INH000001873. J.P. Morgan India Private Limited is also registered with SEBI as a member of the National Stock Exchange of India Limited and the Bombay Stock Exchange Limited (SEBI Registration Number - INZ000239730) and as a Merchant Banker (SEBI Registration Number - MB/INM000002970). Telephone: 91-22-6157 3000, Facsimile: 91-22-6157 3990 and Website: <http://www.jpmpil.com>. JPMorgan Chase Bank, N.A. - Mumbai Branch is licensed by the Reserve Bank of India (RBI) (Licence No. 53/ Licence No. BY.4/94; SEBI - IN/CUS/014/ CDSL : IN-DP-CDSL-444-2008/ IN-DP-NSDL-285-2008/ INBI00000984/ INE231311239) as a Scheduled Commercial Bank in India, which is its primary license allowing it to carry on Banking business in India and other activities, which a Bank branch in India are permitted to undertake. For non-local research material, this material is not distributed in India by J.P. Morgan India Private Limited. Compliance Officer: Ashutosh Sharma; ashutosh.j.sharma@jpmchase.com; +912261575002. Grievance Officer: Ramprasad K, jpmpil.research.feedback@jpmorgan.com; +912261573000. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Please visit [Terms and Conditions and Most Important Terms and Conditions \(MITC\)](#). The annual Compliance audit report is available at <http://www.jpmpil.com/#research>.

Indonesia: PT J.P. Morgan Sekuritas Indonesia is a member of the Indonesia Stock Exchange and is registered and supervised by the Otoritas Jasa Keuangan (OJK).

Korea: J.P. Morgan Securities (Far East) Limited, Seoul Branch, is a member of the Korea Exchange (KRX). JPMorgan Chase Bank, N.A., Seoul Branch, is licensed as a branch office of foreign bank (JPMorgan Chase Bank, N.A.) in Korea. Both entities are regulated by the Financial Services Commission (FSC) and the Financial Supervisory Service (FSS). For non-macro research material, the material is distributed in Korea by or through J.P. Morgan Securities (Far East) Limited, Seoul Branch.

Japan: JPMorgan Securities Japan Co., Ltd. and JPMorgan Chase Bank, N.A., Tokyo Branch are regulated by the Financial Services Agency in Japan.

Malaysia: This material is issued and distributed in Malaysia by JPMorgan Securities (Malaysia) Sdn Bhd (18146-X), which is a Participating Organization of Bursa Malaysia Berhad and holds a Capital Markets Services License issued by the Securities Commission in Malaysia.

Mexico: J.P. Morgan Casa de Bolsa, S.A. de C.V., J.P. Morgan Grupo Financiero is member of the Mexican Stock Exchange ("Bolsa Mexicana de Valores") and the Institutional Stock Exchange ("Bolsa Institucional de Valores"), and it is authorized to act as a broker dealer by the National Banking and Securities Exchange Commission ("Comisión Nacional Bancaria y de Valores").

New Zealand: This material is issued and distributed by JPMSAL in New Zealand only to "wholesale clients" (as defined in the Financial Markets Conduct Act 2013). JPMSAL is registered as a Financial Service Provider under the Financial Service providers (Registration and Dispute Resolution) Act of 2008.

Philippines: J.P. Morgan Securities Philippines Inc. is a Trading Participant of the Philippine Stock Exchange and a member of the Securities Clearing Corporation of the Philippines and the Securities Investor Protection Fund. It is regulated by the Securities and Exchange Commission.

Singapore: This material is issued and distributed in Singapore by or through J.P. Morgan Securities Singapore Private Limited (JPMS) [MDDI (P) 057/08/2025 and Co. Reg. No.: 199405335R], which is a member of the Singapore Exchange Securities Trading Limited, and/or JPMorgan Chase Bank, N.A., Singapore branch (JPMCB Singapore), both of which are regulated by the Monetary Authority of Singapore. This material is issued and distributed in Singapore only to accredited investors, expert investors and institutional investors, as defined in Section 4A of the Securities and Futures Act, Cap. 289 (SFA). This material is not intended to be issued or distributed to any retail investors or any other investors that do not fall into the classes of "accredited investors," "expert investors" or "institutional investors," as defined under Section 4A of the SFA. Recipients of this material in Singapore are to contact JPMS or JPMCB Singapore in respect of any matters arising from, or in connection with, the material.

South Africa: J.P. Morgan Equities South Africa Proprietary Limited and JPMorgan Chase Bank, N.A., Johannesburg Branch are members of the Johannesburg Securities Exchange and are regulated by the Financial Services Conduct Authority (FSCA).

Taiwan: J.P. Morgan Securities (Taiwan) Limited is a participant of the Taiwan Stock Exchange (company-type) and regulated by the Taiwan Securities and Futures Bureau. Material relating to equity securities is issued and distributed in Taiwan by J.P. Morgan Securities (Taiwan) Limited, subject to the license scope and the applicable laws and the regulations in Taiwan. **To the extent that J.P. Morgan Securities (Taiwan) Limited produces research materials on securities not listed on the Taiwan Stock Exchange or Taipei Exchange ("Non-Taiwan Listed Securities"), these materials shall not constitute securities recommendations for the purpose of applicable Taiwan regulations, and, for the avoidance of doubt, J.P. Morgan Securities (Taiwan) Limited does not act as broker for Non-Taiwan Listed Securities.** According to Paragraph 2, Article 7-1 of Operational Regulations Governing Securities Firms Recommending Trades in Securities to Customers (as amended or supplemented) and/or other applicable laws or regulations, please note that the recipient of this material is not permitted to engage in any activities in connection with the material that may give rise to conflicts of interests, unless otherwise disclosed in the "Important Disclosures" in this material.

Thailand: This material is issued and distributed in Thailand by JPMorgan Securities (Thailand) Ltd., which is a member of the Stock Exchange of Thailand and is regulated by the Ministry of Finance and the Securities and Exchange Commission. The registered address is 548 One City Center Building, 50th Floor, Ploenchit Road, Lumpini, Pathum Wan, Bangkok 10330.

UK: Research is produced in the UK by J.P. Morgan Securities plc ("JPMS plc") which is a member of the London Stock Exchange and is authorised by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority or J.P. Morgan Markets Limited ("JPML Ltd") which is authorised and regulated by the Financial Conduct Authority. Unless specified to the contrary, this material is distributed in the UK by JPMS plc and is directed in the UK only to: (a) persons having professional experience in matters relating to investments falling within article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) (Order) 2005 ("the FPO"); (b) persons outlined in article 49 of the FPO (high net worth companies, unincorporated associations or partnerships, the trustees of high value trusts, etc.); or (c) any persons to whom this communication may otherwise lawfully be made; all such persons being referred to as "UK relevant persons". This material must not be acted on or relied on by persons who are not UK relevant persons. Any investment or investment activity to which this material relates is only available to UK relevant persons and will be engaged in only with UK relevant persons. A description of J.P. Morgan EMEA's policy for prevention and avoidance of conflicts of interest related to the production of Research can be found at the following link: [J.P. Morgan EMEA - Research Independence Policy](#).

U.S.: J.P. Morgan Securities LLC ("JPMS") is a member of the NYSE, FINRA, SIPC, and the NFA. JPMorgan Chase Bank, N.A. is a member of the FDIC. Material published by non-U.S. affiliates is distributed in the U.S. by JPMS who accepts responsibility for its content.

General: Additional information is available upon request. The information in this material has been obtained from sources believed to be reliable. While all reasonable care has been taken to ensure that the facts stated in this material are accurate and that the forecasts, opinions and expectations contained herein are fair and reasonable, JPMorgan Chase & Co. or its affiliates and/or subsidiaries (collectively J.P. Morgan) make no representations or warranties whatsoever to the completeness or accuracy of the material provided, except with respect to any disclosures relative to J.P. Morgan and the Research Analyst's involvement with the issuer that is the subject of the material. Accordingly, no reliance should be placed on the accuracy, fairness or completeness of the information contained in this material. There may be certain discrepancies with data and/or limited content in this material as a result of calculations, adjustments, translations to different languages, and/or local regulatory restrictions, as applicable. These discrepancies should not impact the overall investment analysis, views and/or recommendations of the subject company(ies) that may be discussed in the material. Artificial intelligence tools may have been used in the preparation of this material, including assisting in data analysis, pattern recognition, and content drafting for research material. J.P. Morgan accepts no liability whatsoever for any loss

arising from any use of this material or its contents, and neither J.P. Morgan nor any of its respective directors, officers or employees, shall be in any way responsible for the contents hereof, apart from the liabilities and responsibilities that may be imposed on them by the relevant regulatory authority in the jurisdiction in question, or the regulatory regime thereunder. Opinions, forecasts or projections contained in this material represent J.P. Morgan's current opinions or judgment as of the date of the material only and are therefore subject to change without notice. Periodic updates may be provided on companies/industries based on company-specific developments or announcements, market conditions or any other publicly available information. There can be no assurance that future results or events will be consistent with any such opinions, forecasts or projections, which represent only one possible outcome. Furthermore, such opinions, forecasts or projections are subject to certain risks, uncertainties and assumptions that have not been verified, and future actual results or events could differ materially. The value of, or income from, any investments referred to in this material may fluctuate and/or be affected by changes in exchange rates. All pricing is indicative as of the close of market for the securities discussed, unless otherwise stated. Past performance is not indicative of future results. Accordingly, investors may receive back less than originally invested. This material is not intended as an offer or solicitation for the purchase or sale of any financial instrument. The opinions and recommendations herein do not take into account individual client circumstances, objectives, or needs and are not intended as recommendations of particular securities, financial instruments or strategies to particular clients. This material may include views on structured securities, options, futures and other derivatives. These are complex instruments, may involve a high degree of risk and may be appropriate investments only for sophisticated investors who are capable of understanding and assuming the risks involved. The recipients of this material must make their own independent decisions regarding any securities or financial instruments mentioned herein and should seek advice from such independent financial, legal, tax or other adviser as they deem necessary. J.P. Morgan may trade as a principal on the basis of the Research Analysts' views and research, and it may also engage in transactions for its own account or for its clients' accounts in a manner inconsistent with the views taken in this material, and J.P. Morgan is under no obligation to ensure that such other communication is brought to the attention of any recipient of this material. Others within J.P. Morgan, including Strategists, Sales staff and other Research Analysts, may take views that are inconsistent with those taken in this material. Employees of J.P. Morgan not involved in the preparation of this material may have investments in the securities (or derivatives of such securities) mentioned in this material and may trade them in ways different from those discussed in this material. This material is not an advertisement for or marketing of any issuer, its products or services, or its securities in any jurisdiction.

Confidentiality and Security Notice: This transmission may contain information that is privileged, confidential, legally privileged, and/or exempt from disclosure under applicable law. If you are not the intended recipient, you are hereby notified that any disclosure, copying, distribution, or use of the information contained herein (including any reliance thereon) is STRICTLY PROHIBITED. Although this transmission and any attachments are believed to be free of any virus or other defect that might affect any computer system into which it is received and opened, it is the responsibility of the recipient to ensure that it is virus free and no responsibility is accepted by JPMorgan Chase & Co., its subsidiaries and affiliates, as applicable, for any loss or damage arising in any way from its use. If you received this transmission in error, please immediately contact the sender and destroy the material in its entirety, whether in electronic or hard copy format. This message is subject to electronic monitoring: <https://www.jpmorgan.com/disclosures/email>

MSCI: Certain information herein ("Information") is reproduced by permission of MSCI Inc., its affiliates and information providers ("MSCI") ©2026. No reproduction or dissemination of the Information is permitted without an appropriate license. MSCI MAKES NO EXPRESS OR IMPLIED WARRANTIES (INCLUDING MERCHANTABILITY OR FITNESS) AS TO THE INFORMATION AND DISCLAIMS ALL LIABILITY TO THE EXTENT PERMITTED BY LAW. No Information constitutes investment advice, except for any applicable Information from MSCI ESG Research. Subject also to [msci.com/disclaimer](https://www.msci.com/disclaimer)

Sustainalytics: Certain information, data, analyses and opinions contained herein are reproduced by permission of Sustainalytics and: (1) includes the proprietary information of Sustainalytics; (2) may not be copied or redistributed except as specifically authorized; (3) do not constitute investment advice nor an endorsement of any product or project; (4) are provided solely for informational purposes; and (5) are not warranted to be complete, accurate or timely. Sustainalytics is not responsible for any trading decisions, damages or other losses related to it or its use. The use of the data is subject to conditions available at <https://www.sustainalytics.com/legal-disclaimers>. ©2026 Sustainalytics. All Rights Reserved.

"Other Disclosures" last revised July 04, 2026.

Copyright 2026 JPMorgan Chase & Co. All rights reserved. This material or any portion hereof may not be reprinted, sold or redistributed without the written consent of J.P. Morgan. It is strictly prohibited to use or share without prior written consent from J.P. Morgan any research material received from J.P. Morgan or an authorized third-party ("J.P. Morgan Data") in any third-party artificial intelligence ("AI") systems or models when such J.P. Morgan Data is accessible by a third-party.

Completed 14 Aug 2026 03:36 PM JST

Disseminated 14 Aug 2026 03:42 PM JST